SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: R FOR DATA SCIENCE

CAPSTONE PROJECT

```
>
> svm_pred <- predict(svm_model, test_data)
>
> svm_conf <- confusionMatrix(svm_pred, test_data$label)
>
> print("===== SVM Results =====")
[1] "===== SVM Results ====="
> print(svm_conf)
Confusion Matrix and Statistics

          Reference
Prediction ham spam
      ham  553    9
      spam   7   91

               Accuracy : 0.9758
                 95% CI : (0.9609, 0.9861)
    No Information Rate : 0.8485
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9049

 Mcnemar's Test P-Value : 0.8026

            Sensitivity : 0.9875
            Specificity : 0.9100
         Pos Pred Value : 0.9840
         Neg Pred Value : 0.9286
             Prevalence : 0.8485
         Detection Rate : 0.8379
   Detection Prevalence : 0.8515
      Balanced Accuracy : 0.9487

       'Positive' Class : ham

>
> svm_accuracy <- as.numeric(svm_conf$overall["Accuracy"])
>
>
> rf_model <- randomForest(label ~ ., data = train_data)
```

```
> rf_pred <- predict(rf_model, test_data)
>
> rf_conf <- confusionMatrix(rf_pred, test_data$label)
>
> print("===== Random Forest Results =====")
[1] "===== Random Forest Results ====="
> print(rf_conf)
Confusion Matrix and Statistics

          Reference
Prediction ham spam
      ham  559    3
      spam   1   97

               Accuracy : 0.9939
                 95% CI : (0.9846, 0.9983)
    No Information Rate : 0.8485
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9762

 Mcnemar's Test P-Value : 0.6171

            Sensitivity : 0.9982
            Specificity : 0.9700
         Pos Pred Value : 0.9947
         Neg Pred Value : 0.9898
             Prevalence : 0.8485
         Detection Rate : 0.8470
   Detection Prevalence : 0.8515
      Balanced Accuracy : 0.9841

       'Positive' Class : ham

>
> rf_accuracy <- as.numeric(rf_conf$overall["Accuracy"])
>
> accuracy_values <- c(svm_accuracy, rf_accuracy)
> model_names <- c("SVM", "Random Forest")
>
> barplot(accuracy_values,
```

AASHKA MISHRA

NAME: AASHKA MISHRA
ROLL NO: S095

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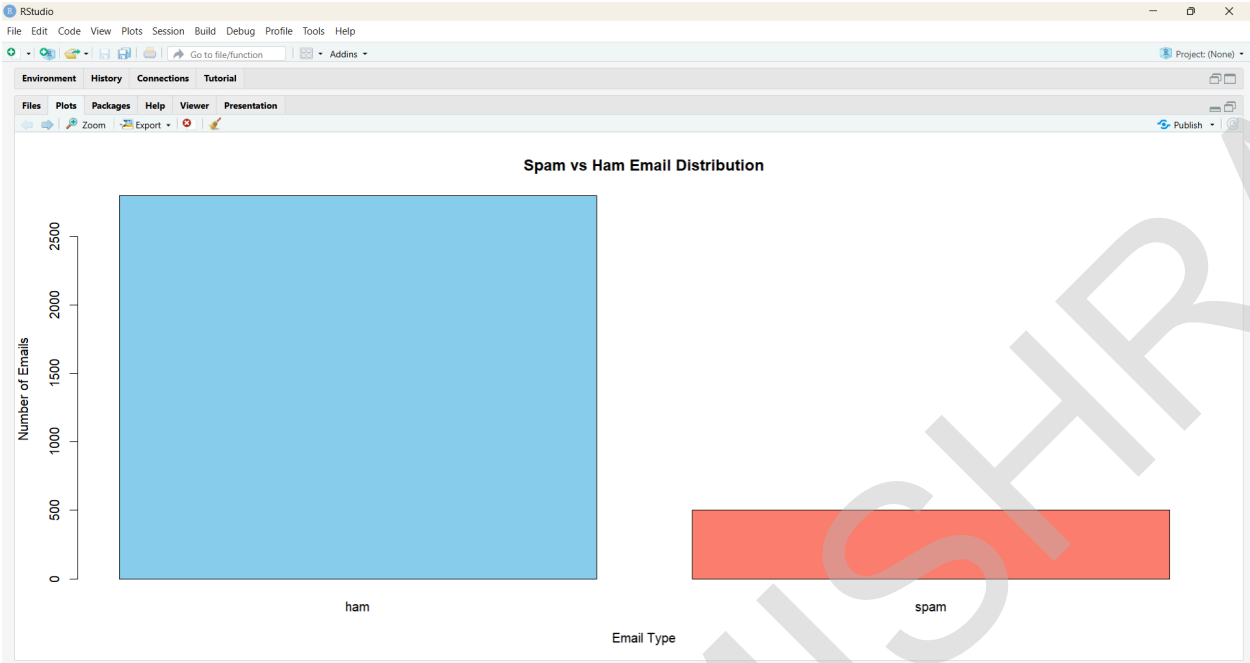

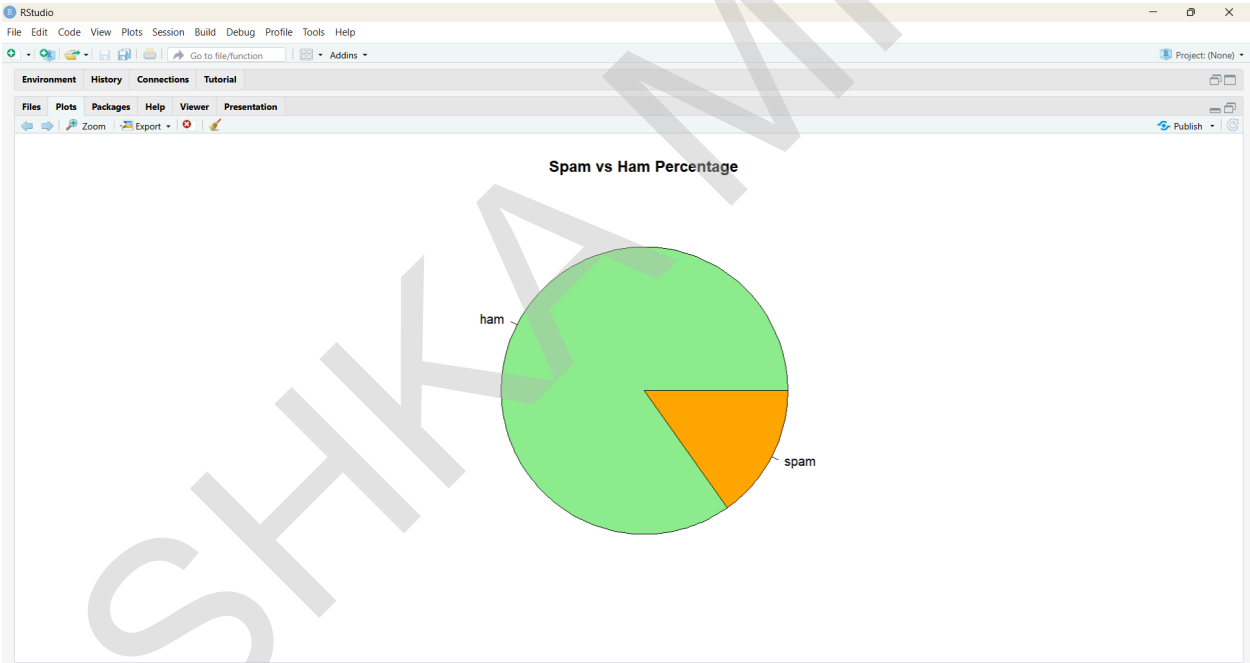

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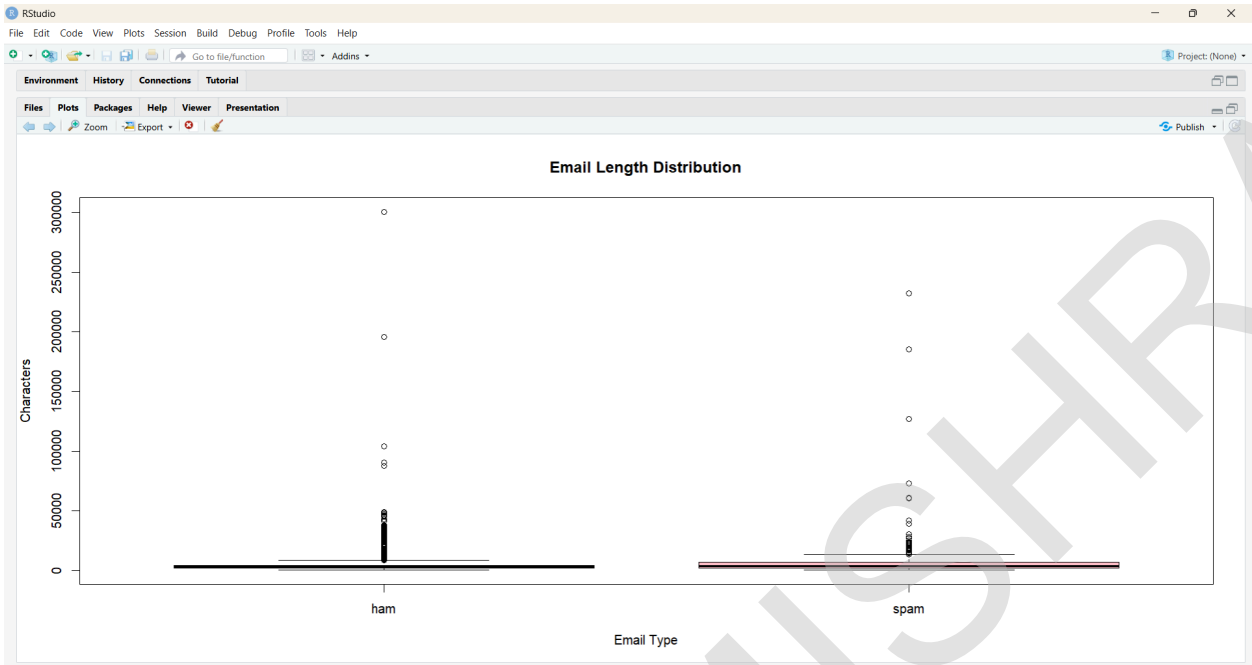

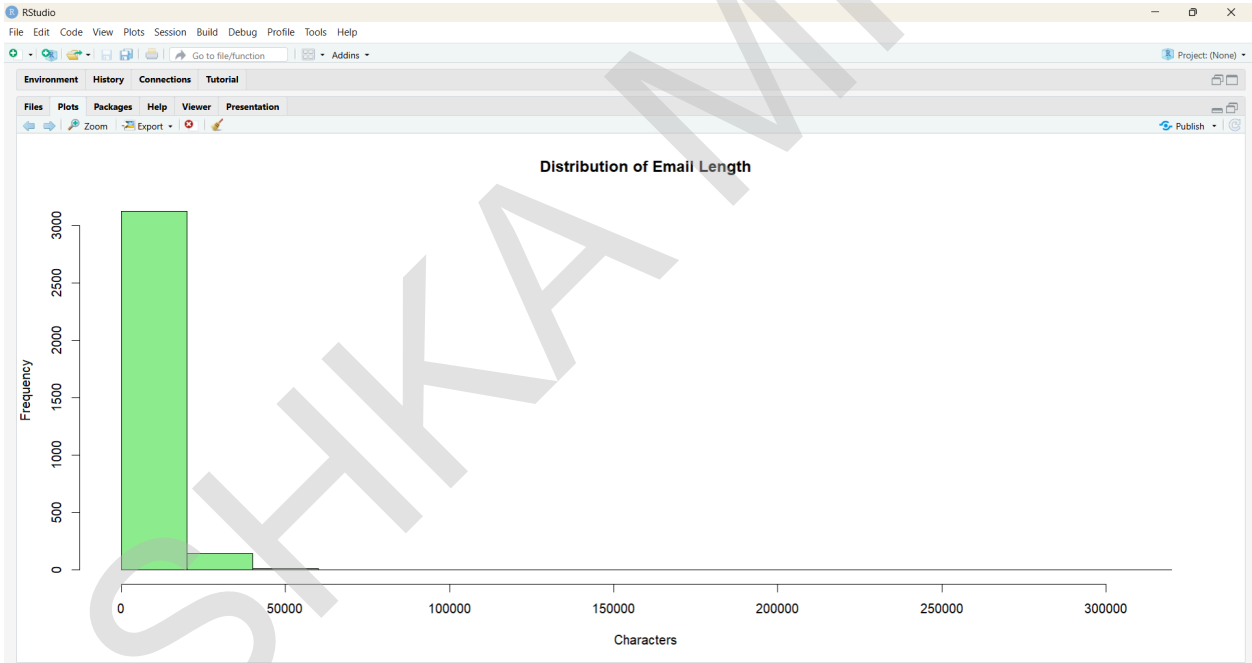

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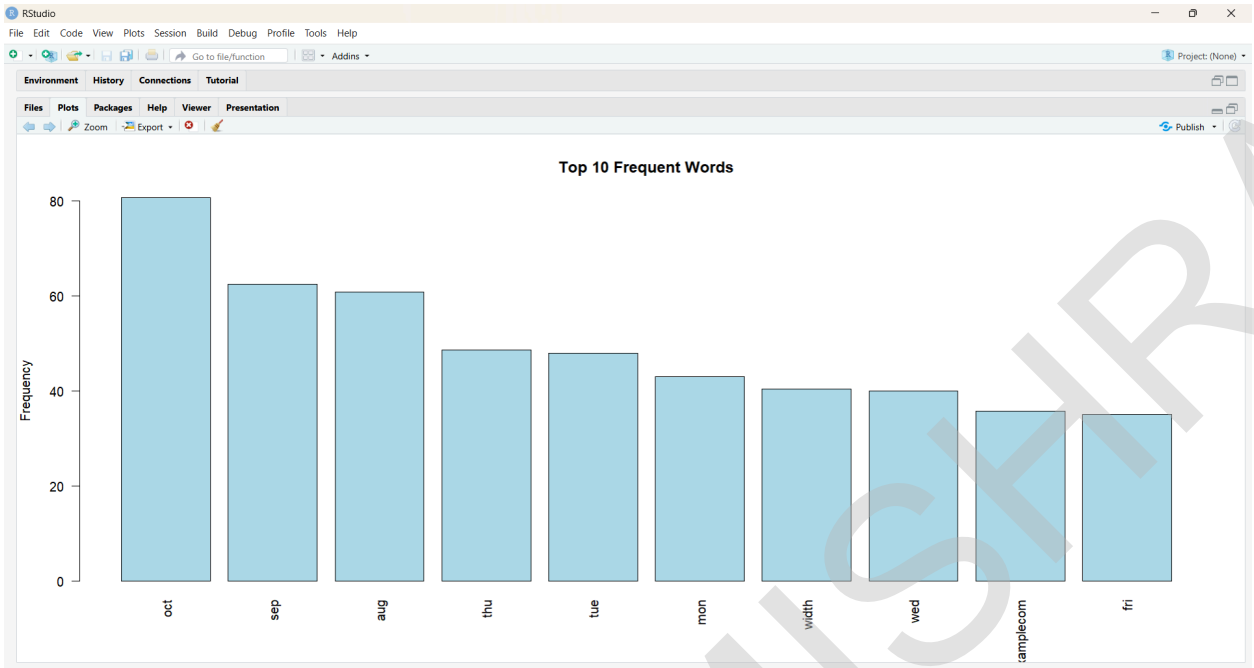

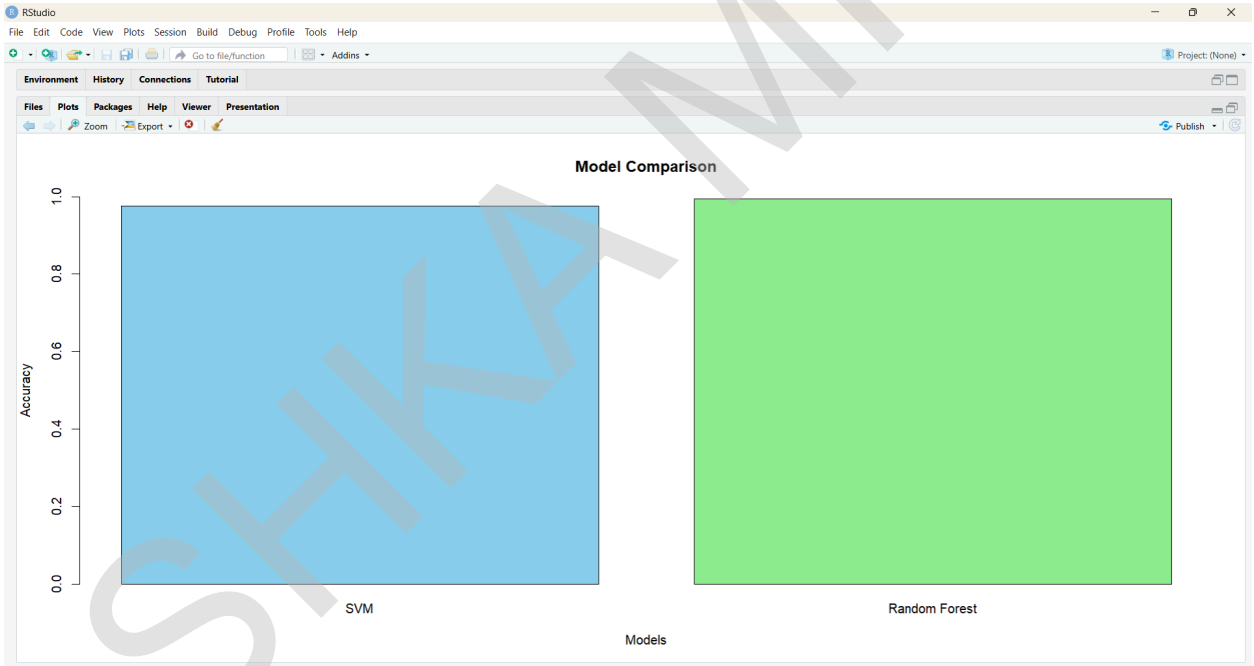

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